

Personal details and date of CV

Surname: Ye **First name:** Boshuai

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Citizenship: Chinese **Date of CV:** August 2026

Degrees

Feb 2025 – present PhD Candidate, Quantum Software Engineering
M3S Research Unit, University of Oulu, Oulu, Finland
Supervisor: Assoc. Prof. Arif Ali Khan, University of Oulu
Co-supervisor: Dr. Marko Paavola, VTT Technical Research Centre of Finland

2023 – Dec 2024 M.Sc. (Tech.), Computer Science
Aalto University, Espoo, Finland
Specialisation: Algorithms, Logic, and Computation; Quantum Technology
Grade: 4.71/5.00 *Honours Degree*

Sep 2019 –Jun 2023 B.Sc., Information Processing Science
University of Oulu, Oulu, Finland
Specialisation: Software Engineering
Grade: 4.02/5.00

Current employment

Feb 2025 – present PhD Candidate
M3S Empirical Software Engineering Research Unit, University of Oulu, Oulu, Finland
Research career stage: Stage I (Doctoral Researcher)
Research focus: reliable quantum software workflows across HPC and heterogeneous computing infrastructures; experimental claim validation under software-visible perturbations; HPC-supported exploration for reliable workflow decision-making.

Previous work experience

Jun 2024 – Feb 2025 Junior Researcher
M3S Empirical Software Engineering Research Unit, University of Oulu, Oulu, Finland
Led end-to-end design and implementation of C2|Q>, a decision-oriented quantum software framework automating the mapping from classical optimisation problems to executable quantum workflows. Defined system abstractions spanning problem modelling, algorithm selection, circuit generation, and hardware-aware execution; evaluated the framework on real quantum toolchains to expose practical execution constraints and variability.

Career breaks

No career breaks to report.

Research funding and grants

2026 Research Grant, Information Technology Research Foundation
(Tietotekniikan tutkimussäätiö), Finland €2,000
Project: *Software Engineering Support for Quantum Workflow Generation and Hardware-Aware Execution*

2026 Research Visit Funding, SafeEarth Visitor Programme, University of Oulu
Research visit support €3,600

2023 – 2025 Full Scholarship, M.Sc. Programme
Aalto University

Research output

Peer-reviewed journal articles and reports

- **B. Ye**, A. A. Khan, T. Pihkakoski, P. Liang, M. A. Akbar, M. Silveri, L. Malmi. C2|Q>: A Robust Framework for Bridging Classical and Quantum Software Development. *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2026.
[doi:10.1145/3803018](https://doi.org/10.1145/3803018)
- **B. Ye**, A. A. Khan, T. Pihkakoski, P. Liang, M. A. Akbar, M. Silveri, L. Malmi. C2|Q>: A Robust Framework for Bridging Classical and Quantum Software Development (RCR Report). *ACM Transactions on Software Engineering and Methodology (TOSEM)*, published 3 July 2026. [doi:10.1145/3833408](https://doi.org/10.1145/3833408)

Peer-reviewed conference papers

- **B. Ye**, A. A. Khan, P. Liang. When T-Depth Misleads: Predicting Fault-Tolerant Quantum Execution Slowdown under Magic-State Delivery Constraints. *IEEE International Conference on Quantum Computing and Engineering (QCE)*, accepted, 2026.
[arXiv:2604.11409](https://arxiv.org/abs/2604.11409)
- M. A. Akbar, S. Rafi, A. A. Khan, **B. Ye**. Quantum Secure DevOps (QSecOps): Integrating Quantum-Based Security Checks into CI/CD Pipelines: Next-Generation Software Security. *ACM International Conference on the Foundations of Software Engineering (FSE Companion)*, 2025. [doi:10.1145/3696630.3731624](https://doi.org/10.1145/3696630.3731624)
- **B. Ye**. C2|Q>: Unifying Classical and Quantum Programming. *ACM/SIGAPP Symposium on Applied Computing (SAC)*, Student Research Competition, 2025.
[doi:10.1145/3672608.3707991](https://doi.org/10.1145/3672608.3707991)
- M. Esposito, M. T. Sabzevari, **B. Ye**, D. Falessi, A. A. Khan, D. Taibi. Classi|Q>: Towards a Translation Framework to Bridge the Classical-Quantum Programming Gap. *ACM International Workshop on Quantum Software (Q-SE)*, 2024.

Manuscripts under review

- **B. Ye**, P. Liang, M. T. Sabzevari, A. A. Khan. Auditing Empirical Comparisons in Quantum Software. Submitted to the *IEEE/ACM International Conference on Software Engineering (ICSE 2027)*, 2026. [arXiv:2607.00516](https://arxiv.org/abs/2607.00516)
- **B. Ye** et al. Beyond Equivalence: Synthesis Availability Across Quantum Compilation Boundaries. Submitted to the *ACM International Conference on the Foundations of Software Engineering (FSE 2027)*, Research Papers Track, 2026.

Preprints

- A. A. Khan, **B. Ye**, M. A. Akbar, J. A. Khan, D. Mougouei, X. Ma. Mining Q&A Platforms for Empirical Evidence on Quantum Software Programming. [arXiv:2503.05240](https://arxiv.org/abs/2503.05240), 2025.
[arXiv:2503.05240](https://arxiv.org/abs/2503.05240)

Software and research artifacts

- **C2|Q** — modular framework translating classical problem specifications into executable hybrid quantum workflows. github.com/C2-Q/C2Q
- **ClaimStab-QC** — source-bounded research framework and artifact for auditing the

stability of empirical comparative claims in quantum software through explicit evidence records, locked audit designs, and scoped stability outcomes.

github.com/Bossy-Ye/ClaimStab-QC

Research supervision and leadership experience

Master's thesis supervision support

University of Oulu

Contributed to the supervision of Sang Vo's master's thesis: "*Quantum-Enhanced CNN for Highly Accurate Breast Cancer Classification: A Proof-of-Concept Study*." Provided guidance on quantum software engineering aspects, quantum machine learning concepts, experimental design, and implementation-level issues.

Teaching merits

Teaching assistant, Next Generation of Quantum Software Engineering

University of

Oulu, Spring 2025 and Spring 2026

Supported course activities related to quantum software engineering, including student guidance, technical support on quantum software tooling, and assistance with exercises and project work.

Teaching Lead, Quantum Software Engineering Summer School

University of Oulu,

1–15 August 2026

Led lectures, Q&A sessions, and technical discussions on quantum computing, quantum software tools, algorithms, and development workflows.

Invited lecturer, Quantum Software Engineering course

LUT University, 2025

Delivered invited teaching on quantum software engineering, with a focus on the basics of quantum computing, quantum programming tools, software engineering challenges, and practical development workflows.

Invited speaker, Quantum Software Engineering workshop

LUT University, 2026

Awards and honours

2025 Master's Degree with Honours, Aalto University

2022 Honours Student, University of Oulu

Other key academic merits

Reviewer: Served as a reviewer for the ACM International Workshop on Quantum Software Engineering (Q-SE).

Open science: Primary research artifacts (C2|Q), ClaimStab-QC, Quantum-Q&A-Mining) are publicly archived on GitHub with documented reproducibility instructions. Research manuscripts are deposited on arXiv when appropriate to support open access and reproducibility.

Scientific and societal impact

Research contributes to the reliability and reproducibility of quantum software results, directly relevant to the trustworthiness of quantum computing in scientific and industrial applications. The C2|Q framework is developed within a Business Finland-supported programme targeting practical quantum software tools for industry and research.

ClaimStab-QC develops a systematic approach for auditing whether empirical comparative claims remain supported within explicit, reproducible evidence boundaries.

Other

Technical skills: Qiskit, PennyLane; OpenQASM and circuit IRs; quantum transpilation and noise-aware compilation; hybrid classical–quantum programming; fault-tolerant resource estimation; Python, C/C++; Linux, Git, Docker, CI/CD.

Languages: Chinese (native), English (professional working proficiency).